

THE UNIVERSITY OF DODOMA
COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INTERNET PROGRAMMING AND APPLICATION I
PRACTICAL TEST (30 MIN)

Question One

A private secondary school called Mwanga Academy is developing a student performance dashboard using jQuery, Bootstrap 4, and plain JavaScript. The dashboard page loads student data and presents it as an interactive, searchable table.

Given the following array of student result objects:

```
var results = [  
  { name:"Zawadi Kimani",   regNo:"S001", math:78, english:85,  
    science:90 },  
  { name:"Brian Ochieng",   regNo:"S002", math:45, english:52,  
    science:60 },  
  { name:"Nadia Salum",     regNo:"S003", math:92, english:88,  
    science:95 },  
  { name:"Peter Luwasa",    regNo:"S004", math:33, english:41,  
    science:38 },  
  { name:"Sylvia Mhando",   regNo:"S005", math:67, english:71,  
    science:74 }  
];
```

- a) Using jQuery, write code that populates a Bootstrap table (`#resultsTable > tbody#tableBody`) on page load. Each row must show: Name, Reg No, Math, English, Science, Average (calculated as the mean of the three subjects, rounded to 1 decimal place), and Grade. Apply Bootstrap's `table-success` class to rows where average ≥ 70 , `table-warning` where average is 50–69, and `table-danger` where average < 50 .
- b) Implement a live search on an input (`#searchInput`) using the input event: hide rows where neither the student name nor registration number matches the typed text (case-insensitive). Show a `#noResults` paragraph if no rows are visible.
- c) Add a "Sort by Average" button (`#btnSort`). Each click must re-sort the results array by average (descending) and re-render the entire table body using jQuery. Maintain the Bootstrap row-colour rules after each sort.